

Improving interface-mediated carrier transport for efficient perovskite/Cu(In,Ga)Se₂ tandem solar cells

Received: 13 November 2025

Accepted: 22 July 2026

Published online: 05 August 2026

 Check for updates

Li Zeng^{1,7}, Wuji Wang^{1,7}, Liting Tang^{1,7}, Xing Gao¹, Haofei Xia², Xiaomin Wang²✉, Zekai Luo¹, Yuqi Zhao¹, Yan Gao¹, Zhuo Xue¹, Wenbo Li¹, Yue Sun¹, Jiangtong Zhao³, Jiajv Wang¹, Yong Liu¹, Weiqing Zhou^{1,4}, Rui Sun³, Jie Min³, Guilin Chen⁵✉, Ti Wang¹, Sheng Wang¹, Shengjun Yuan^{1,4,6}, Xudong Xiao¹✉ & Jianmin Li¹✉

Monolithic perovskite/Cu(In,Ga)Se₂ tandem solar cells can surpass the efficiency limits of single-junction photovoltaics but are currently limited by interfacial optical and electronic losses. Here we demonstrate a dual-interface carrier transport engineering strategy that combines a nanoparticle-assisted NiO_x intermediate recombination layer with bimolecular co-passivation at the perovskite/C₆₀ interface, enabling conformal coverage on textured Cu(In,Ga)Se₂ surfaces, suppressed defects, optimized band alignment and enhanced carrier extraction across both interfaces. Champion monolithic tandem cells achieve power conversion efficiencies of 31.09% (certified 30.57%, steady-state 30.32%) for small-area devices (0.0539 cm²) and 29.44% (certified 28.85%) for larger-area devices (1.0298 cm²). The optimized devices retain ~94% of their initial efficiency after >3,500 h of storage, ~91% after >750 h of continuous operation, and ~90% after 960 h of heating at 70 °C, demonstrating simultaneously enhanced efficiency and stability.

Tandem solar cells (TSCs) offer a transformative approach to surpassing the Shockley–Queisser efficiency limit of single-junction devices¹. Among emerging materials, perovskites have attracted considerable attention due to their tunable bandgaps and exceptional optoelectronic properties, enabling high-performance tandem architectures such as perovskite/silicon (Si)^{2,3}, all-perovskite^{4–6} and perovskite/Cu(In,Ga)Se₂ (CIGS)^{7–9}. Perovskite/CIGS TSCs are particularly attractive because the CIGS bottom cell can be engineered to ~1.00 eV for optimal spectral utilization¹⁰, while exhibiting radiation hardness, operational stability and compatibility with flexible substrates¹¹.

Despite these advantages, achieving high-efficiency monolithic perovskite/CIGS tandems remains challenging. Key barriers include

suboptimal energy-level alignment between subcells, non-ideal intermediate recombination layers (IRLs) for rough CIGS surfaces, difficulties in forming high-quality perovskite films, and inefficient carrier transport across interfaces¹¹. Recent works have emphasized the critical role of IRLs in two-terminal (2T) devices. Chemical–mechanical polishing was initially used to reduce shunting, achieving power conversion efficiencies (PCEs) of up to 22.4%¹². Later work introduced NiO_x/poly[bis(4-phenyl)(2,4,6-trimethylphenyl)amine] (PTAA) bilayers¹³ or self-assembled monolayers (SAMs) such as [2-(3,6-dimethoxy-9H-carbazol-9-yl)ethyl]phosphonic acid (MeO-2PACz)¹⁴ and NiO_x/SAM bilayers⁹ to establish conformal coverage and energy-aligned interfaces without polishing.

¹Key Lab of Artificial Micro- and Nano-structures of Ministry of Education, School of Physics and Technology, Wuhan University, Wuhan, China. ²Hubei Key Laboratory of Plasma Chemistry and Advanced Materials, School of Materials Science and Engineering, Wuhan Institute of Technology, Wuhan, China.

³The Institute for Advanced Studies, Wuhan University, Wuhan, China. ⁴Wuhan Institute of Quantum Technology, Wuhan, China. ⁵College of Physics and Energy, Fujian Normal University, Fuzhou, China. ⁶School of Artificial Intelligence, Wuhan University, Wuhan, China. ⁷These authors contributed equally: Li Zeng, Wuji Wang, Liting Tang. ✉e-mail: wxm@wit.edu.cn; glchen@fjnu.edu.cn; xdxiao@whu.edu.cn; ljmphy@whu.edu.cn

While the above strategies improved the device performance, limitations persist: sputtered NiO_x (SU- NiO_x) contains substantial defect densities, leading to non-negligible fill-factor (FF) losses^{15–17}, and widely used commercial SAMs such as [4-(3,6-dimethyl-9H-carbazol-9-yl)butyl]phosphonic acid (Me-4PACz) provide insufficient chemical bonding strength with metal oxides and incomplete coverage on textured surfaces, impeding uniform perovskite deposition and efficient carrier extraction¹⁸. Moreover, the bromine (Br)-rich component in the top absorber layer of wide-bandgap (WBG) perovskite (1.65–1.80 eV)¹¹ in 2T TSCs intrinsically introduces substantial bulk and interfacial defects, as well as energy-level misalignment at the perovskite/ C_{60} interface. Such limitations promote non-radiative recombination and reduce carrier transport efficiency, ultimately depressing open-circuit voltage (V_{oc}) and FF^{19,20}. Addressing these coupled challenges at both the bottom (IRL/SAM/perovskite) and top (perovskite/ C_{60}) interfaces in perovskite subcells are thus essential for realizing high-efficiency 2T perovskite/CIGS TSCs.

In this work, we develop a dual-interface carrier transport engineering (DICTE) strategy. At the bottom interface, a bilayer NiO_x hole transport layer (SS- NiO_x) enables conformal coverage over rough CIGS substrates and promotes high-quality perovskite growth, thereby suppressing interfacial recombination and enhancing hole extraction. At the top interface, a bimolecular co-passivation (BMCP) system passivates defects, reduces non-radiative recombination and optimizes energy-level alignment with C_{60} , facilitating efficient carrier extraction. As a result, monolithic perovskite/CIGS tandems achieve a certified efficiency of 30.57% (steady-state 30.32%) for small-area (0.0539 cm²) devices and 28.85% for larger-area (1.0298 cm²) devices, with champion efficiencies of 31.09% and 29.44%, respectively. The devices also exhibit excellent stability, retaining 94.38% of their initial efficiency after 3,576 h in storage, 91.49% after 756 h of maximum power point tracking (MPPT) and 90.55% after more than 960 h of thermal stress (70 °C) without encapsulation.

Conformal bilayer NiO_x hole transport structure

To mitigate potential shunting caused by the textured CIGS surface, we proposed a NiO_x nanoparticles (NiO_x NPs)-assisted double-layer hole transport structure (denoted as SS- NiO_x). As illustrated schematically in Fig. 1a, the SS- NiO_x is fabricated by sequentially sputtering a NiO_x layer, followed by spin-coating a thin NiO_x NPs (SI- NiO_x) layer, and finally crystallizing the stack through air annealing. For comparison, three IRLs (SU- NiO_x , SI- NiO_x and SS- NiO_x) were systematically investigated on the bottom-cell surface. Systematic studies reveal that SU- NiO_x (~15 nm; Supplementary Figs. 1–4 and Supplementary Note 1) suffers from shunting when thinner and parasitic absorption when thicker (Supplementary Fig. 5), while SI- NiO_x (10 mg ml⁻¹; Supplementary Fig. 6) may provide incomplete coverage, resulting in relatively low efficiency and widely scattered V_{oc} distribution (Supplementary Fig. 7 and Supplementary Note 2). By contrast, the SS- NiO_x structure achieves balanced coverage and favourable photovoltaic properties, delivering the highest PCE with enhanced FF and without sacrificing V_{oc} (Fig. 1b and Supplementary Fig. 8), as indicated by a reduction in reverse leakage current from 1.61×10^{-3} to 8.42×10^{-7} mA cm⁻² in dark-state current density–voltage (J – V) curves (Supplementary Fig. 8d), indicating suppressed non-radiative recombination.

Scanning electron microscopy (SEM) analysis (Fig. 1c) reveals distinct coverage behaviours of perovskite films on CIGS with different NiO_x interlayers. SU- NiO_x leaves sharp Al-doped ZnO (AZO) tips exposed, indicating poor coverage. SI- NiO_x improves macroscopic coverage but still permits micrometre-sized particles. By contrast, SS- NiO_x achieves conformal coverage with minimal voids, yielding the most uniform film. Corresponding experiments on CIGS/AZO substrates confirm that nanoparticle assistance is essential for continuous film formation of perovskite (Fig. 1c, insets). Detailed microstructural characterization (Fig. 1d, Supplementary Figs. 9 and 10

and Supplementary Notes 3 and 4) and energy-dispersive X-ray (EDX) mapping (Supplementary Fig. 11 and Supplementary Note 5) show that NiO_x NPs fill concave regions on micrometre-scale features of the CIGS surface, while SI- NiO_x alone fails to form conformal coverage. This is further supported by time-of-flight secondary ion mass spectrometry (TOF-SIMS) and X-ray photoelectron spectroscopy (XPS) analyses (Supplementary Figs. 12 and 13). The conformal nanoparticle layer reduces substrate roughness (r.m.s. from 84.1 to 76.0 nm; Supplementary Fig. 14), facilitating perovskite deposition and resulting in films with lower surface roughness (Supplementary Fig. 15) and more compact, uniformly sized grains (Supplementary Fig. 16) compared with controls, highlighting substrate-mediated regulation of perovskite crystallization and improved film quality.

Consistently, X-ray diffraction (XRD) analysis (Fig. 1e) shows that NiO_x NPs enhance the (100):(110) peak intensity ratio of perovskite by a factor of 4.1–5.6 relative to SU- NiO_x , indicating a stronger [100] orientation that favours vertical charge transport^{21,22}. The perovskite (100)/lead iodide (PbI_2) peak ratio also increases markedly from 4.25 (SU- NiO_x) to 27.53 (SS- NiO_x), indicating suppressed PbI_2 precipitation (Supplementary Fig. 17). Because PbI_2 clusters act as non-radiative recombination centres and interfacial barriers, their reduction improves charge separation and extraction^{23,24}.

XPS and Raman characterizations (Fig. 1f,g and Supplementary Figs. 18–20) reveal distinct surface chemistry among the NiO_x interlayers. In the O 1s spectra, SU- NiO_x shows a high –OH content (34.4%) due to sputtering-induced oxygen vacancies^{16,25}, while SS- NiO_x exhibits reduced –OH (21.7%) and increased Ni^{3+} –O (21.0%) arising from annealing-induced formation of NiOOH . These changes enhance film conductivity (Supplementary Table 1) and provide surface –OOH groups for SAM attachment, facilitating uniform perovskite deposition^{15,26}. Ni 2p spectra show that the $\text{Ni}^{3+}/\text{Ni}^{2+}$ ratio increases from 2.11 (SI- NiO_x) to 2.75 (SU- NiO_x) and 2.86 (SS- NiO_x), with the Ni 2p_{3/2} peak shifting from 853.7 eV (SI- NiO_x) to 854.2 eV (SS- NiO_x), indicating modified surface electronic environments that favour hole transport. Raman spectra further confirm the increase of Ni^{3+} , indicated by weakened 2LO intensity (1,056.4 cm⁻¹) in SS- NiO_x (ref. 27). Together, these results demonstrate that SS- NiO_x provides the optimal surface chemistry to support SAM and perovskite growth.

As a result, SAM deposition on different NiO_x surfaces reveals clear variations in surface wetting and binding behaviour. NiO_x NPs slightly improve the wettability of SAM films (contact angle from 13.1° to 10.7°) (Supplementary Fig. 21). XPS and EDX mapping analyses (Supplementary Figs. 22–24, Supplementary Note 6 and Supplementary Table 2) reveal sparse phosphorus (P) regions on SU- NiO_x , whereas SI- and SS- NiO_x exhibited denser SAM coverage, accompanied by smoother surface morphology as indicated by atomic force microscopy (AFM) measurements (Supplementary Fig. 25). Reflection Fourier transform infrared spectroscopy (Supplementary Fig. 26) further confirms the change in the binding effect of PA–OH interactions through shifted P–OH absorption peaks²⁸. After SAM modification, SS- NiO_x exhibits better wettability for perovskite precursor solutions than SU- NiO_x , with contact angles reduced from 32.8° to 23.1° (Supplementary Fig. 27), attributed to abundant surface hydroxyl groups and homogeneous SAM anchoring. This optimized interface promotes subsequent perovskite growth.

Interfacial charge extraction mechanisms

To elucidate the mechanisms underlying improved device performance, we conducted a series of interfacial and charge transport characterizations (Fig. 2). Kelvin probe force microscopy (KPFM) reveals that SS- NiO_x exhibits the lowest average surface potential (ASP) of ~290 mV and the narrowest FWHM of 37.8 mV, indicating both an increased work function (W_F) and superior spatial uniformity compared with SU- and SI- NiO_x (Fig. 2a–c, Supplementary Fig. 28 and Supplementary Note 7). Ultraviolet (UV) photoelectron spectroscopy (UPS; Fig. 2d–f,

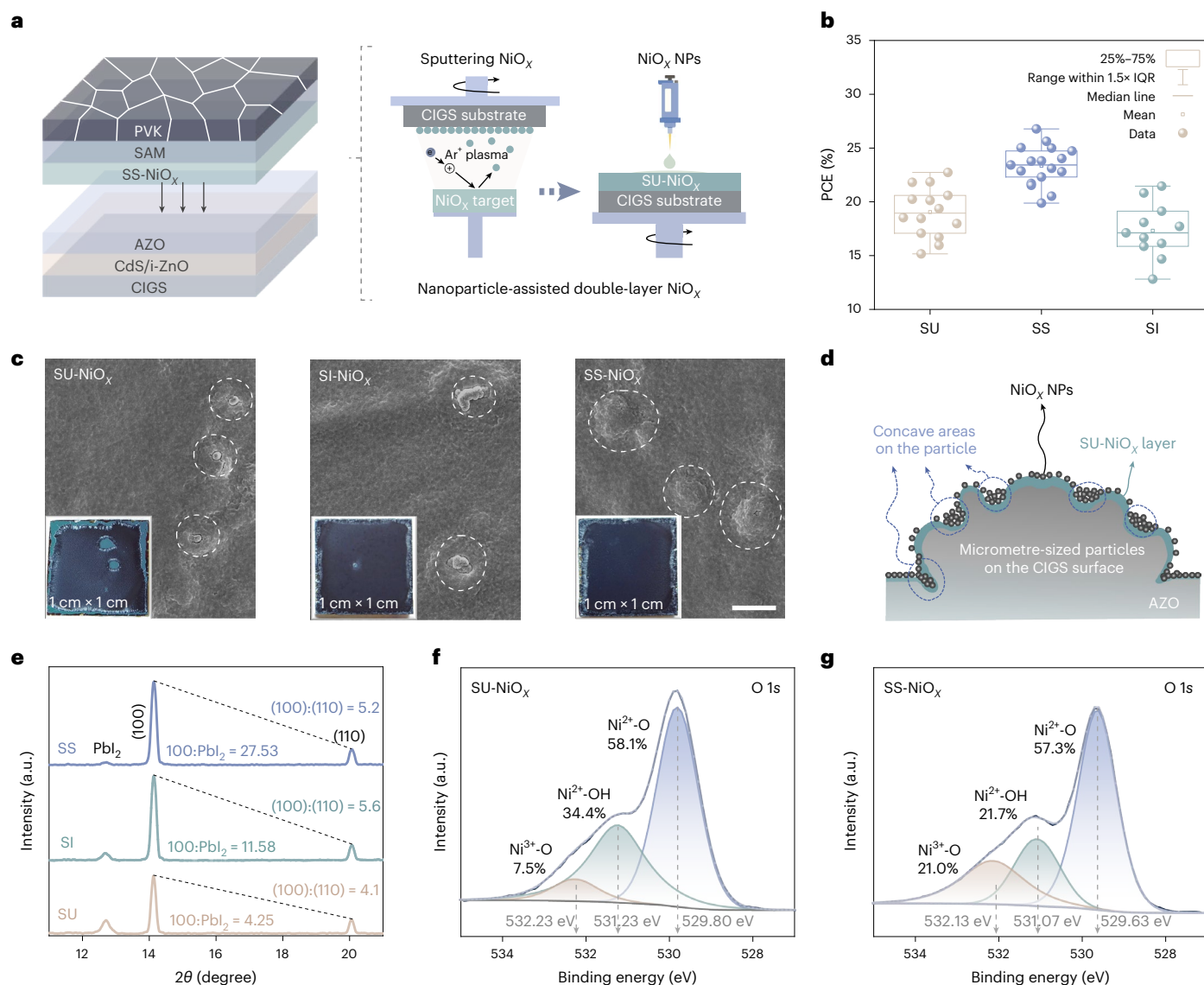


Fig. 1 | Conformal NiO_x coverage and dense perovskite growth. **a**, Schematic diagram of the SS-NiO_x preparation process in perovskite/CIGS tandem devices. **b**, Statistical distributions of PCEs for perovskite/CIGS tandem devices with SU-NiO_x, SS-NiO_x and SI-NiO_x. Each data point corresponds to an independent measurement, with sample sizes of $n = 14$, 17 and 11, respectively. The box boundaries, minimum, maximum, interquartile range (IQR), median and mean of the box plots are defined in the legend. **c**, Surface SEM images of perovskite films

on CIGS bottom cells with SU-NiO_x, SI-NiO_x and SS-NiO_x. Scale bar, 3 μm. Insets show photographs of the corresponding CIGS/AZO substrates. **d**, Schematic diagram of SS-NiO_x deposited on a micrometre-sized particle. **e**, Enlarged XRD spectra of perovskite films on SU-NiO_x, SI-NiO_x and SS-NiO_x in the 2θ range of 11°–21°, with corresponding intensity ratios of (100)/PbI₂ and (100)/(110). **f**, O 1s XPS spectra of SU-NiO_x. **g**, O 1s XPS spectra of SS-NiO_x films.

Supplementary Fig. 29 and Supplementary Note 8) further shows that SS-NiO_x reduces the valence band offset (ΔE_{VBM}) with Me-4PACz from 0.24 eV (SU-NiO_x) to 0.18 eV, and aligns the Fermi level more favourably ($\Delta E^* = +0.25$ eV), effectively minimizing hole transport barriers and interfacial energy losses, facilitating selective hole extraction while blocking electrons. These combined effects optimize the charge extraction pathway from perovskite to the NiO_x/SAM interface, improving V_{OC} , FF and short-circuit current density (J_{SC}) in tandem devices.

Electrical measurements of AZO/NiO_x/SAM/Au devices indicate that SS-NiO_x exhibits the steepest current rise under forward bias, reflecting minimal resistance and rapid hole transport (Fig. 2g). Hall-effect measurements confirm increased carrier concentration and reduced sheet resistance (Supplementary Table 1), while electrochemical impedance spectroscopy shows a higher recombination resistance (R_{rec}) at the SS-NiO_x interface (Supplementary Fig. 30 and Supplementary Table 3), indicative of fewer defect traps and enhanced

carrier transport. These improvements are further quantified by space-charge-limited current measurements on hole-only devices (Supplementary Fig. 31 and Supplementary Note 9).

Photoluminescence (PL) and time-resolved PL (TRPL) of buried perovskite interfaces on glass/AZO/NiO_x/SAM stacks reveal pronounced PL quenching and faster TRPL decay for SS-NiO_x, with the average lifetime decreasing from 305.2 ns (SU-NiO_x) to 97.6 ns (SS-NiO_x) (Fig. 2h,i and Supplementary Table 4), indicating enhanced hole transport and decrease in non-radiative recombination²⁹. A concomitant blue shift from 760 nm to 749 nm further reveals improved passivation at buried interfaces²⁷. Transient absorption (TA) spectroscopy corroborates accelerated ground-state bleach decay (5.3 ns versus 8.9 ns) and decreased band-gap renormalization (BGR) energy (71.04 meV versus 82.10 meV) (Supplementary Figs. 32–34 and Supplementary Table 5), confirming enhanced interfacial charge transfer and reduced defect density²⁰. Overall, the double-layer NiO_x design plays a critical role in

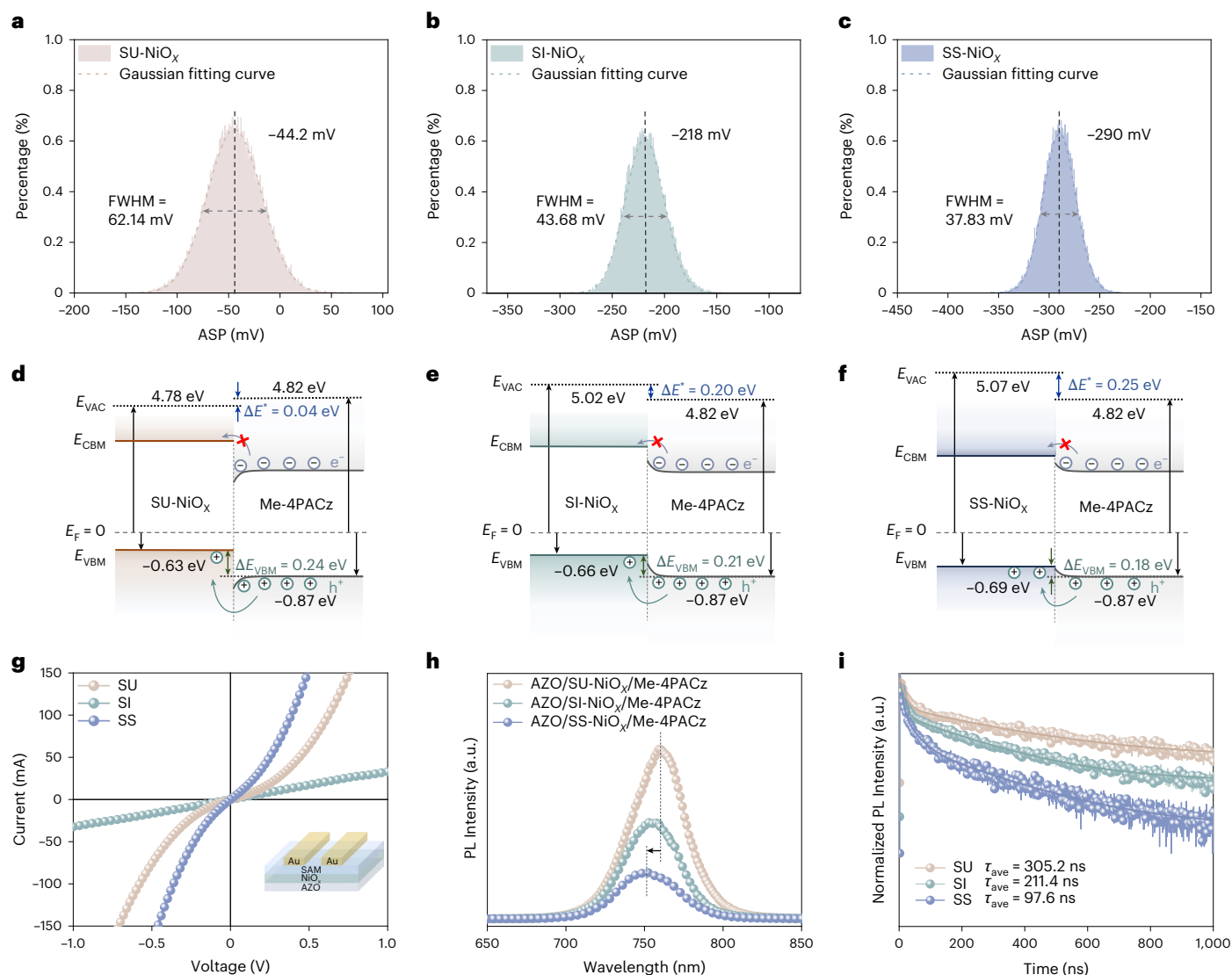


Fig. 2 | Energy levels and carrier transport in different NiO_x films.

a–c, Statistical distributions of surface potential, along with the corresponding Gaussian fitting curves for SU- (**a**), SI- (**b**) and SS-NiO_x (**c**) films. **d–f**, Interfacial energy-level diagrams for SU- (**d**), SI- (**e**) and SS-NiO_x (**f**) and Me-4PACz (values for

Me-4PACz from ref. 35). **g**, Conductivity current density–voltage ($J-V$) curves of ‘AZO/NiO_x/SAM/Au’ devices. **h**, PL spectra of perovskite films on AZO with different NiO_x layers. **i**, TRPL decay and bi-exponential fitting curves of perovskite films on SU-, SI- and SS-NiO_x substrates.

facilitating charge transport across the perovskite/CIGS interface, laying the foundation for the reliable fabrication of high-performance tandem devices.

Synergistic passivation of perovskite

To further enhance the performance of the WBG perovskite top cell, we developed a BMCP strategy by combining ethylenediamine diiodide (EDADI) and 4-fluorophenylethylammonium iodide (p-F-PEAI) additives via a mixed-solvent approach, followed by one-step spin-coating onto the crystallized perovskite layer (Fig. 3a and Supplementary Fig. 35). The perovskite subcells are based on a glass/ITO/SI-NiO_x/SAM/WBG-perovskite/C₆₀/BCP/Ag architecture (where ITO is indium tin oxide) with the chemical formula of Cs_{0.05}(FA_{0.83}MA_{0.17})_{0.95}Pb(I_{0.8}Br_{0.2})₃ and a bandgap of ~1.65 eV (Supplementary Fig. 36). After perovskite crystallization (control), we applied spin-coating of EDADI (E), p-F-PEAI (P), sequential EDADI then p-F-PEAI (EP), p-F-PEAI then EDADI (PE) and their mixed solution (BMCP). All treatments except the PE sequence improved device performance relative to control, with BMCP yielding the highest PCE (Fig. 3b and Supplementary Fig. 37).

Defects in the perovskite film act as non-radiative recombination centres, impairing carrier transport. XPS measurements show increased I 3d binding energy for all passivation films, suggesting a modified chemical environment of iodine (Fig. 3c). Density functional theory (DFT) calculations on a representative FAPbI₃ (100) surface reveal that the I[−] ions from the passivation molecules can effectively passivate iodine vacancy defects (V_I). Meanwhile, the NH₃⁺ groups of EDADI²⁺ and p-F-PEAI⁺ are likely to interact with cationic interstitial regions and bind with iodine, further suppressing V_I (Supplementary Figs. 38 and 39 and Supplementary Note 10). Moreover, EDADI²⁺ and p-F-PEAI⁺ can fill the vacancies of Pb²⁺ and FA⁺, respectively, thereby repairing the [PbI₆]^{4−} framework to passivate cationic defects (Fig. 3d and Supplementary Figs. 40–42). The combined application simultaneously passivates anionic and cationic defects, with a calculated binding energy of ~3.16 eV (Fig. 3e) indicating strong supramolecular stabilization and enhanced surface anchoring, which are expected to reduce non-radiative recombination.

SEM and AFM characterization of perovskite films reveal distinct morphological changes induced by different passivation treatments (Fig. 3f). The unpassivated control films exhibit small grains, pinholes

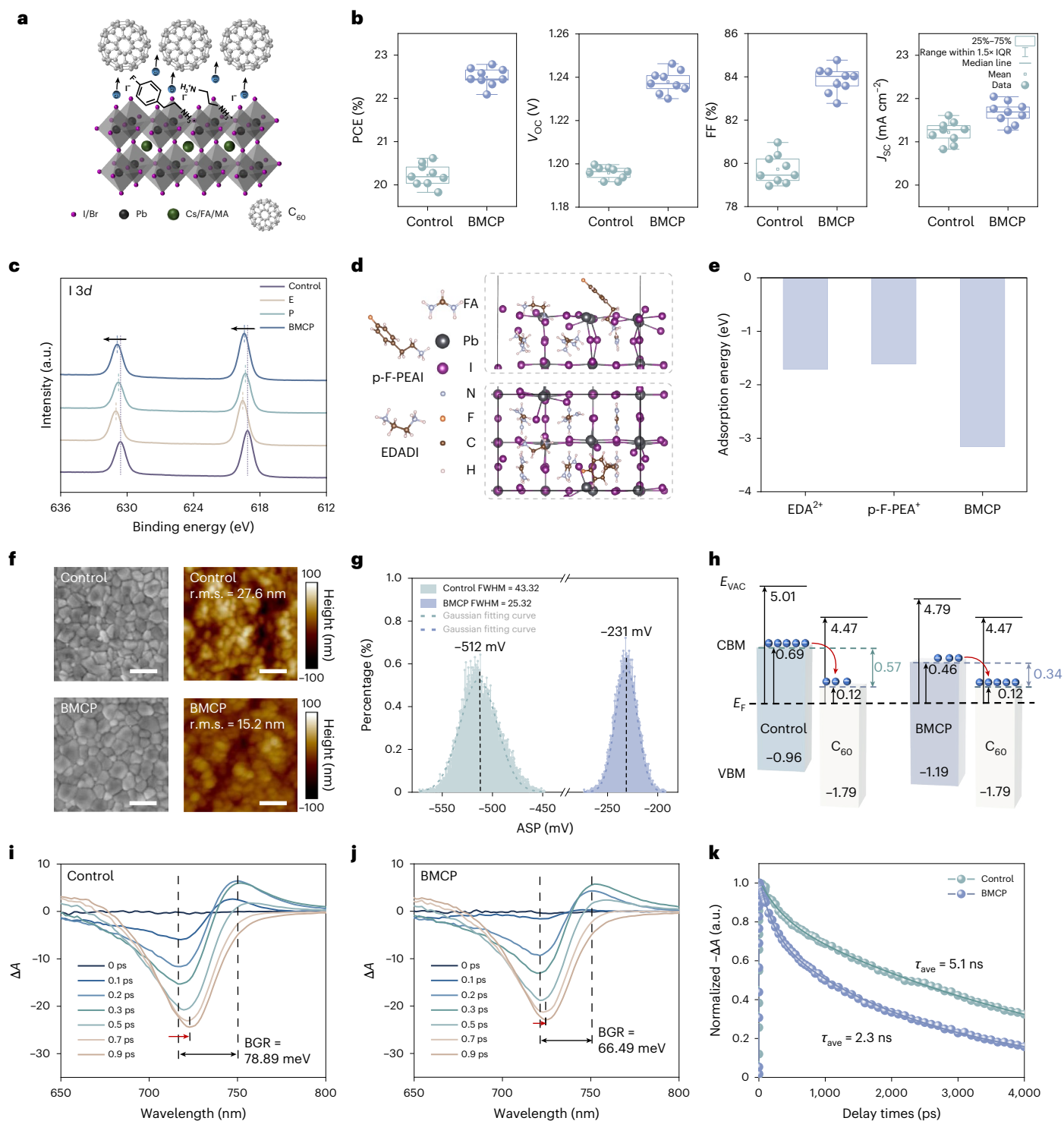


Fig. 3 | Passivation and carrier transport at the perovskite/C₆₀ interface.

a, Schematic illustration of the BMCP strategy applied to perovskite films. **b**, Statistical distributions of photovoltaic parameters for the control and BMCP-modified PSCs. Each data point corresponds to a statistical set ($n = 9$). The box boundaries, minimum, maximum, IQR, median and mean of the box plots are defined in the legend. **c**, XPS spectra of the I 3d of perovskite films with and without different passivation modes. **d**, Relaxed configurations of the BMCP-modified adsorption model on the FAPbI₃ (100) perovskite surface in side (top) and top (bottom) views. **e**, Calculated adsorption energy between ammonium

cation and defective FAPbI₃ (100) surfaces. **f**, SEM (left) and AFM (right) images comparing control and BMCP-treated perovskite films. Scale bar, 500 nm. **g**, Potential signal distribution and fitting curves of perovskite films with and without BMCP treatment. **h**, Energy-level alignment of control and BMCP-treated films relative to C₆₀. CBM, conduction band minimum; VBM, valence band maximum. **i, j**, TA spectra (0–0.9 ps) of control (i) and BMCP-treated (j) perovskite films. BGR, band-gap renormalization. **k**, Temporal decay dynamics and bi-exponential fitting curves of the prominent photobleaching signals in perovskite films with and without BMCP treatment.

and high surface roughness, whereas BMCP treatment films produce uniformly enlarged grains with smooth surfaces (Supplementary Figs. 43–45 and Supplementary Note 11). Root-mean-square (r.m.s.) roughness decreases from 27.6 nm (control) to 15.2 nm (BMCP), indicating that passivation in general smooths the film and reduces defect density, with BMCP achieving optimal surface uniformity. XRD analysis further confirms improved crystallinity after BMCP treatment, with the (100)/PbI₂ peak intensity ratio increasing from 2.23 (control) to 3.30 and a narrower full width at half maximum (FWHM) at both the (100) and (210) planes (Supplementary Fig. 46). These results indicate reduced PbI₂ segregation, with potentially lower defect density, and prolonged carrier lifetimes³⁰. Consistently, BMCP films exhibit the strongest PL intensity, reflecting enhanced photon emission efficiency and crystal quality (Supplementary Fig. 47), along with reduced reverse saturation dark current (Supplementary Fig. 48).

To investigate the interface-modifying effect of the BMCP strategy, W_f simulations were performed (Supplementary Fig. 49 and Supplementary Note 12), showing that EDADI primarily contributes to band modulation and substantially alters the perovskite surface W_f . KPFM and UPS measurements were then used to analyse the effect of these molecules on band alignment. As shown in Fig. 3g, BMCP-treated perovskite films exhibit a higher ASP (−231 mV versus −512 mV for control) and reduced surface potential fluctuations, indicating increased W_f and improved spatial homogeneity, which collectively suppress interfacial recombination and enhance charge transport (Supplementary Fig. 50 and Supplementary Note 13). As shown in Fig. 3h and Supplementary Fig. 51, the treatment also reduces W_f from 5.01 eV to 4.79 eV and shifts the Fermi-level upwards, weakening surface p -type character and lowering minority carrier concentration. At the perovskite/C₆₀ interface, the conduction band offset decreases from 0.57 eV to 0.34 eV, facilitating faster electron extraction and less interfacial recombination³¹. Conductive AFM (C-AFM) (Supplementary Fig. 52 and Supplementary Note 14) confirms improved surface conductivity with forward photocurrent rising from 6.6 pA to 16.4 pA (ref. 32), while electrochemical impedance spectroscopy analysis shows an increased R_{rec} from 7,900 Ω to 21,059 Ω (Supplementary Fig. 53 and Supplementary Table 6), further indicating suppressed interfacial recombination and enhanced charge transport. Furthermore, the light intensity-dependent V_{oc} (Suns- V_{oc}) measurements reveal that BMCP-treated devices have a lower ideality factor ($n = 1.24$) than the control ($n = 1.47$), confirming suppression of deep-level trap-mediated non-radiative recombination. In addition, FF loss analysis shows that BMCP reduces charge transfer losses from 7.1% to 5.3% compared with the control (Supplementary Fig. 54 and Supplementary Note 15).

To probe the impact of BMCP-induced band structure modifications on recombination and carrier transport at the perovskite/C₆₀ interface, TA, transient photovoltage (TPV) and transient photocurrent (TPC) measurements were conducted. TA two-dimensional plots (Supplementary Fig. 55) reveal a GSB band at ~723 nm with rapidly decaying dynamics and an excited-state absorption signal near 750.5 nm, with a slight redshift (red arrow) attributed to band filling and BGR. BMCP-treated films exhibit reduced BGR energy (66.49 meV versus 78.89 meV for control; Fig. 3i,j), indicating lower defect density and background carrier concentration. Relaxation dynamics (Fig. 3k and Supplementary Table 7) show a decreased average lifetime from 5.1 ns to 2.3 ns, reflecting enhanced carrier extraction via surface defect passivation and optimized interfacial energy alignment. TPV and TPC analyses (Supplementary Fig. 56) further confirm improved carrier lifetimes and extraction dynamics. The carrier lifetime based on TPV increases from 1.99 μ s to 4.55 μ s, while photocurrent decay decreases from 0.55 μ s to 0.27 μ s, indicating suppressed non-radiative recombination and accelerated charge extraction. Consequently, the BMCP strategy enables perovskite solar cells (PSCs) with excellent performance, accompanied by enhanced V_{oc} and FF arising from reduced

trap density, optimized energy levels and efficient charge transport, collectively promoting high-efficiency TSCs.

Performance and stability of perovskite/CIGS tandems

Integration of the BMCP strategy and SS-NiO_x improves the quality of both intermediate and top interfaces. This DICTE strategy enables the fabrication of high-performance perovskite/CIGS tandem devices (Fig. 4a,b). Devices prepared using this strategy, denoted SSD, exhibit performance enhancements comparable to single-junction perovskite cells. The ideality factor (n) of the tandem devices decreases from 2.39 to 1.82 (Supplementary Fig. 57), indicating suppressed non-radiative recombination³³. Spatially resolved PL lifetime mapping via confocal fluorescence lifetime imaging microscopy (FLIM) reveals accelerated carrier extraction at both interfaces: among the glass/AZO/NiO_x/SAM/perovskite structures, SSD tandems exhibit the shortest PL lifetime, indicating faster carrier transport through the AZO layer, while in the glass/AZO/NiO_x/SAM/perovskite/C₆₀ structures, rapid PL quenching confirms enhanced electron extraction at the C₆₀ interface (Supplementary Fig. 58). These observations highlight the critical role of dual-interface engineering in minimizing recombination and optimizing charge extraction, resulting in devices with high FF and V_{oc} (Fig. 4b). By further optimizing the thickness and processing parameters of the SS-NiO_x layer and the passivation layer (Supplementary Figs. 59–69), we successfully integrated high-performance monolithic perovskite/CIGS TSCs on home-made CIGS substrates (Supplementary Fig. 70, Supplementary Note 16 and Supplementary Table 8). With aperture masking, a 0.0539-cm² active area device reached a champion PCE of 31.09% with V_{oc} of 1.765 V, FF of 83.11% and J_{sc} of 21.19 mA cm^{−2} (Fig. 4c). Scribing was used to exclude possible contributions from adjacent cells, and statistical histograms confirm the efficiency measurements (Supplementary Fig. 71 and Supplementary Note 17). Upon storage in a glovebox, V_{oc} further increased (Supplementary Fig. 72), approaching the theoretical limit for sub-cell voltage summation (Supplementary Fig. 73 and Supplementary Note 18). The strategy also translated to larger-area devices (1.0298 cm²), which delivered PCE of 29.44%, V_{oc} of 1.828 V, FF of 78.05% and J_{sc} of 20.63 mA cm^{−2} (Fig. 4d and Supplementary Figs. 74 and 75), demonstrating excellent scalability and robust photovoltaic performance. At the same time, the external quantum efficiency (EQE) spectrum confirms a near-ideal bandgap and current matching between the sub-cells, which was critical to achieving a final PCE exceeding 31% (Fig. 4e and Supplementary Fig. 76).

The best-performing device, cross-certified by the National Institute of Measurement and Testing Technology (NIMTT, China) with an illumination mask aperture of 0.0539 cm², achieved a PCE of 30.57%, V_{oc} of 1.751 V, J_{sc} of 21.24 mA cm^{−2} and FF of 82.22% (Fig. 4f and Supplementary Fig. 77). After 5 min of continuous illumination, the small area cell also reached a steady-state certified efficiency of 30.32% (Supplementary Fig. 77). We further noticed that the shading mask introduced additional lateral resistance in non-illuminated regions. Systematic variation of mask dimensions reveals that voltage and current losses increase as the aperture decreases (larger shaded area), as quantified in Supplementary Figs. 78 and 79, Supplementary Note 19 and Supplementary Table 9. For the larger-area device (aperture mask of 1.0298 cm²), the certified PCE is 28.85% (V_{oc} 1.813 V, J_{sc} 20.91 mA cm^{−2}, FF 76.10%; Fig. 4g and Supplementary Fig. 80), with the FF reduction attributed to higher series resistance in the transparent conductive oxide layer from extended lateral current paths, combined with less efficient carrier extraction at the metal grid contacts. As shown in Fig. 4h and Supplementary Table 10, this work represents one of the highest reported performances for 2T perovskite/CIGS TSCs, highlighting the effectiveness of the proposed interfacial design.

Beyond PCE, long-term durability, thermal stability and light tolerance are key factors for the practical deployment of perovskite/

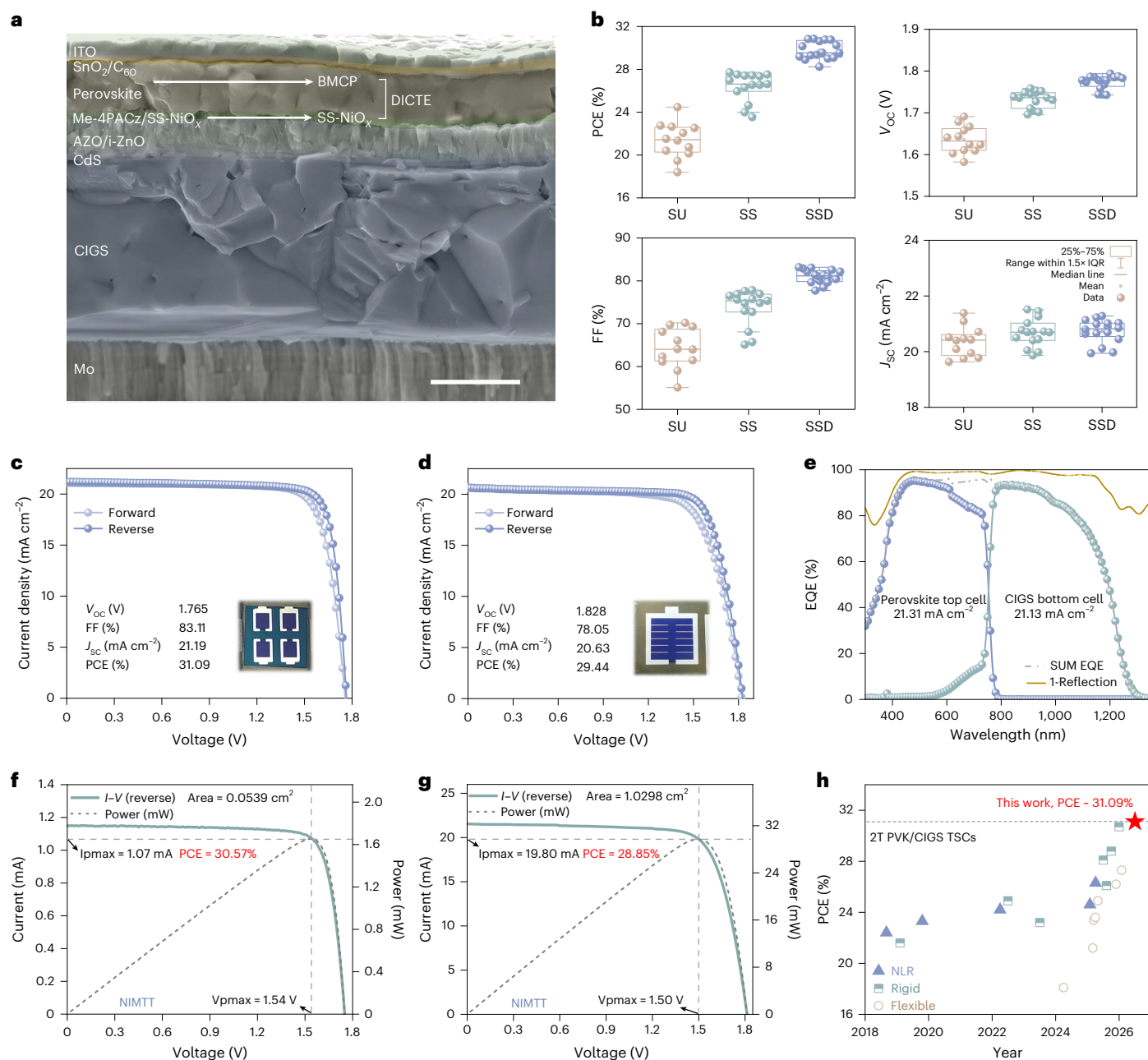


Fig. 4 | Performance of monolithic perovskite/CIGS tandem devices.

a, Structure diagram of monolithic perovskite/CIGS TSCs with the DICTE strategy. Scale bar, 1 μm . **b**, Statistical distributions of photovoltaic parameters for the SU, SS and SSD TSCs. Each data point corresponds to an independent measurement, with sample sizes of $n = 12, 15$ and 16 , respectively. The box boundaries, minimum, maximum, IQR, median and mean of the box plots are defined in the legend. **c**, Laboratory-measured $J-V$ curves of a small-area champion perovskite/CIGS tandem device with an aperture area of 0.0539 cm^2 (inset: photograph of a small-area device). **d**, Laboratory-measured $J-V$ curves

of a larger-area tandem device with an aperture area of 1.0298 cm^2 (inset: photograph of a larger-area device). **e**, EQE curves and total absorbance ($1 - R$) spectrum of the perovskite/CIGS tandem device. SUM, summation. **f**, Current–voltage ($I-V$) curve of the champion tandem device cross-certified by NIMTT (China) for small-area (0.0539 cm^2 aperture area). I_{pmax} and V_{pmax} are the current and voltage at the maximum power point, respectively. **g**, $I-V$ curve of the champion tandem device cross-certified by NIMTT (China) for larger area (1.0298 cm^2 aperture area). **h**, PCE progress of published 2T perovskite/CIGS TSCs in recent years^{7–9,12–14,17,36–47}. NLR, National Laboratory of the Rockies.

CIGS TSCs. Unencapsulated tandem devices employing the DICTE strategy exhibit excellent stability when stored in N_2 , retaining 94.38% of their initial efficiency after 149 days ($\sim 3,576\text{ h}$) at 25°C , based on the average of 17 independent devices, with FF, V_{oc} and J_{sc} maintained at $\sim 95\%$ and $>99\%$ of their initial values, respectively (Fig. 5a). Thermal stability tests in a nitrogen (N_2) glovebox revealed that unencapsulated SU- and SSD-based devices retained over 90% of their initial efficiency after 960 h at 70°C , markedly outperforming control devices, which retained only 72.62% (Fig. 5b). The optimized device remains

chemically inert after ageing, with no appreciable ion interdiffusion or degradation aside from slight PbI_2 formation on the perovskite surface (Supplementary Fig. 81). Statistical operational stability under continuous 1-sun illumination with MPPT also favours SSD devices. After 756 h, control devices retained an average of only 28.86% of their initial PCE, whereas DICTE-modulated SSD tandem devices maintained 91.49% of their original efficiency (Fig. 5c and Supplementary Fig. 82). This enhanced stability may arise from reduced defect density and improved hydrophobicity. Air exposure induces moisture uptake and cation

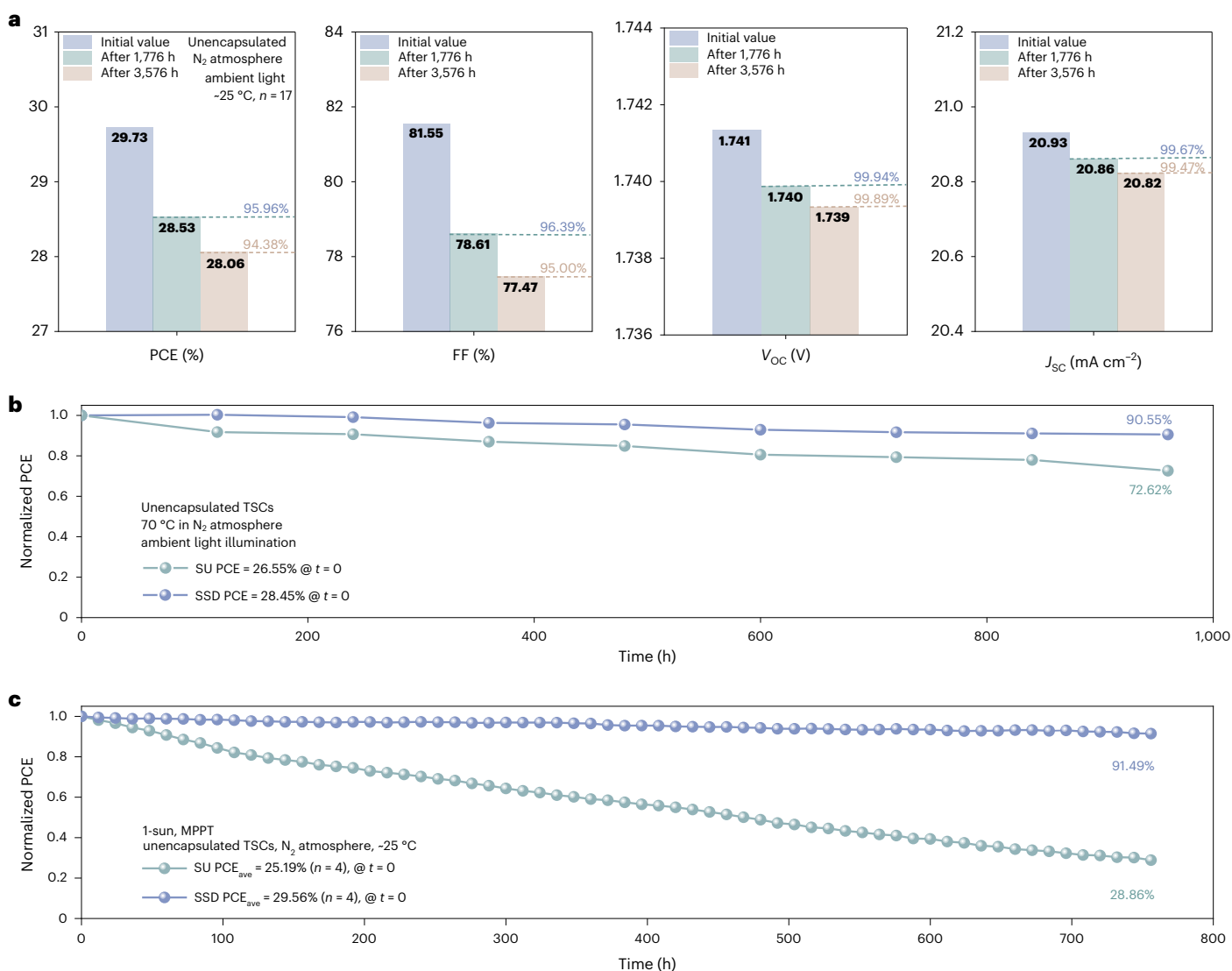


Fig. 5 | Stability of tandem devices. a, Stability of the unencapsulated DICTE-treated perovskite/CIGS tandem devices ($n = 17$) stored in an N_2 atmosphere at room temperature (-25°C) under ambient light illumination with an aperture mask of $\sim 0.0408\text{ cm}^2$. The initial average PCE of the tandem device was 29.73%. **b**, Thermal stability test result of control and target groups at 70°C under ambient light illumination. The initial PCEs of the SU and SSD tandem devices

were 26.55% and 28.45%, respectively. **c**, The average MPPT stability test results of unencapsulated control and target perovskite/CIGS TSCs under continuous 1-sun illumination in an N_2 atmosphere at room temperature (-25°C). The initial average PCEs of the SU and SSD tandem devices were 25.19% and 29.56%, respectively.

deprotonation in perovskite, leading to irreversible decomposition and reduced long-term stability³⁴. As shown in Supplementary Fig. 83 and 84 and Supplementary Note 20, BMCP treatment increases the water contact angle of the perovskite film from 67.6° to 74.2° , which could help suppress moisture-induced degradation.

Conclusions

Achieving high efficiency in monolithic perovskite/CIGS TSCs critically depends on careful engineering of subcell interfaces, particularly for rough and irregularly textured CIGS surfaces. We demonstrate a DICTE strategy that simultaneously targets the bottom and top interfaces of the perovskite absorbers. Nanoparticle-assisted SS- NiO_x deposition at the bottom interface ensures conformal coverage, enhances SAM anchoring and promotes high-quality perovskite crystallization, while molecular co-passivation at the top interface reduces defects and non-radiative recombination at the perovskite/ C_{60} interface. These synergistic modifications optimize energy-level alignment and enable efficient carrier transport across both interfaces. As a result, monolithic

tandem cells achieve a certified efficiency of 30.57% and maintain excellent long-term operational stability. This work underscores the central role of interface engineering in advancing tandem photovoltaics and may provide a generalizable strategy for the design of next-generation high-performance devices.

Methods

Materials

ITO/glass (sheet resistance of $15\ \Omega$ per square), Spiro-OMeTAD and NiO_x NPs (99.999%) were purchased from Advanced Election Technology. Caesium iodide (CsI), lead bromide (PbBr_2), formamidinium iodide (FAI), lead iodide (PbI_2), BCP, EDADl and p-F-PEAI were obtained from Xi'an Polymer Light Technology. Methylammonium bromide (MABr) came from Greatcell Solar Materials Pty. Me-4PACz from TCI. C_{60} , *N,N*-dimethylformamide (DMF), dimethyl sulfoxide (DMSO), ethyl acetate (EA) and isopropyl alcohol (IPA) were purchased from Sigma-Aldrich. The ITO target ($\text{In}_2\text{O}_3/\text{SnO}_2 = 90:10\text{ wt}\%$) and NiO_x target (99.9%) were purchased from Huizhou Tianyi Rare Materials.

Atomic-layer-deposited SnO_2 precursor (99.9999%) was bought from Nanjing Aimuyuan Scientific Equipment. Raw materials for preparing the CIGS absorber were purchased from ZhongNuo Advanced Material Technology and Kurt J. Lesker Company. The reagents mainly used for depositing the cadmium sulfide (CdS) buffer layer are thiourea and cadmium sulfate, which were purchased from Sigma-Aldrich. Molybdenum (Mo), intrinsic zinc oxide (i-ZnO) and AZO targets were also supplied by Huizhou Tianyi Rare Materials. The aluminium (Al) was obtained from Beijing Zhongcheng New Materials Technology, while Au (99.999%) and Ag (99.99%) were purchased from Hebei Fengming New Material Technology.

Precursor solution preparation

The 1.6 M WBG perovskite precursor solution with a target composition of $\text{Cs}_{0.05}(\text{FA}_{0.83}\text{MA}_{0.17})_{0.95}\text{Pb}(\text{I}_{0.8}\text{Br}_{0.2})_3$ was prepared by dissolving 20.8 mg CsI , 30.5 mg MABr_2 , 144.5 mg of PbBr_2 , 228.4 mg of FAI and 607.8 mg of PbI_2 in 1 ml DMF:DMSO (4:1 v/v). Preparation of hole transport layer solutions: 5 mg ml^{-1} NiO_x NPs in deionized H_2O :IPA (3:1 v/v); 1 mg ml^{-1} Me-4PACz in IPA. Other solutions: 1 mg ml^{-1} EDADI in IPA (passivation solution); 20 mg ml^{-1} p-F-PEAI in DMF:DMSO (4:1 v/v, additive-A); 1 mg ml^{-1} p-F-PEAI in EA (additive-B); 3 mg ml^{-1} p-F-PEAI in DMF (buried solution).

Fabrication of 1.65-eV bandgap PSCs

Single-junction perovskite cell structure was given in the main text. ITO glass substrates ($1.5\text{ cm} \times 1.5\text{ cm}$) were ultrasonically cleaned in detergent, deionized water, acetone, IPA and ethanol (30 min each), followed by drying under N_2 . Before spin-coating, the substrates were UV-treated at 100°C for 20–30 min. For the preparation of NiO_x /Me-4PACz double hole-transport layers (HTLs), the NiO_x NP solution was sonicated in an ice-water bath for 20 min and filtered (0.22 μm polytetrafluoroethylene). Then, it was spin-coated onto the substrate at 3,000 rpm for 30 s and subsequently annealed at 120°C in air for 20 min. Substrates were then transferred to an N_2 glove box, followed by the Me-4PACz solution prepared with the same spin-coating process and annealed at 100°C for 10 min. The subsequent buried solution was spin-coated at 4,000 rpm for 20 s. To prepare perovskite films, 1 ml perovskite solution (with 50 μl additive-A) was filtered through a 0.22- μm polytetrafluoroethylene membrane before deposition, then 35 μl was dropped and spin-coated at 4,000 rpm for 35 s. At 10 s before the end of the spin-coating step, 120 μl EA (with additive-B diluted to 0.01 mg ml^{-1}) was dropped onto the substrate. Then, the films were annealed at 100°C for 20 min. For E-treatment, 0.25 mg ml^{-1} EDADI was spin-coated at 4,000 rpm for 60 s and annealed at 55°C for 5 min; for P-treatment, 0.08 mg ml^{-1} p-F-PEAI was spin-coated at 4,000 rpm for 35 s and annealed at 60°C for 10 min; for BMCP-treatment, 0.2 mg ml^{-1} EDADI and 0.05 mg ml^{-1} p-F-PEAI were diluted in a mixed solution of IPA:EA (3:1 v/v) after filtration and shaken for 60 s to mix thoroughly. Then, 60 μl of postprocessing solution was spin-coated onto the perovskite films at 4,000 rpm for 35 s and annealed at 55°C for 5 min. After the perovskite film was prepared, 20-nm C_{60} , 7-nm BCP and 150-nm Ag electrodes were deposited by electron beam evaporation at $0.1\text{ \AA s}^{-1}$ under $-5 \times 10^{-4}\text{ Pa}$.

Fabrication of CIGS solar cells

The CIGS cells we prepared follow the basic structure of glass/ Mo /CIGS/ CdS / i-ZnO /AZO/ Al-grid (Supplementary Fig. 70 and Supplementary Note 16). The manufacturing sequence initiates with the deposition of a 1- μm -thick Mo back contact onto soda-lime glass substrates via d.c. sputtering. Subsequently, the CIGS absorber layer ($\sim 2.5\text{ }\mu\text{m}$) is made by a three-stage co-evaporation methodology. A 60-nm CdS buffer layer is then deposited via the chemical bath deposition method, followed by radiofrequency (RF) sputtered 50-nm i-ZnO and 200-nm AZO. For discrete device implementation, an Al grid contact is ultimately fabricated via an electron-beam thermal evaporation system.

TSC fabrication

The tandem devices were fabricated by first depositing $\sim 15\text{ nm}$ NiO_x on the CIGS cells via RF sputtering at 60 W under an Ar pressure of 0.12 Pa, which is a step not required for single-junction PSCs. The subsequent process, from the deposition of NiO_x NPs to the completion of the perovskite film, was identical to that of single-junction PSCs. To reduce parasitic absorption, only 9 nm of C_{60} was deposited. Furthermore, 20 nm of SnO_2 was deposited by reactive atomic-layer deposition (Picosun R200, Finland) to replace the BCP layer in the single-junction cells. Next, approximately 50 nm of ITO was deposited by RF sputtering at 0.15 Pa as the top transparent conductive oxide of the tandem device, followed by thermal evaporation of a 500 nm silver (Ag) electrode for repeated testing. To reduce optical reflection and improve current density, a $\sim 105\text{-nm}$ MgF_2 anti-reflection layer was deposited by electron beam evaporation at a rate of $0.5\text{ \AA s}^{-1}$.

J–V characterization

The J – V curves were recorded under ambient conditions using an AM 1.5 G solar simulator (Enlitech AAA solar simulator, SS109 X100R). A Keithley 2400 sourcemeter swept voltage from -0.1 V to 1.27 V for single-junction devices and from -0.1 V to 1.85 V for tandem devices (0.1 V s^{-1} scan speed, 20 mV voltage steps and 1 ms delay time). A certified Si reference solar cell (Enli Tech. Optoelectronic Calibration Lab) was used to calibrate the light intensity to 100 mW cm^{-2} . For the monolithic perovskite/CIGS tandem devices, polyethylene naphthalate masks with different aperture areas, including 0.0539 cm^2 , were used during measurements.

Stability test

The MPPT stability measurement was conducted under continuous 1-sun illumination provided by an LED solar simulator (SLS-LED-80, Qingdao Solar Scientific Instrument High-tech), with the stabilized outputs of the photovoltaic parameters monitored near the maximum power point.

EQE measurement

The EQE was determined with a QE/incident photon-to-electron conversion efficiency system (Enlitech, QE-R666) by focusing monochromatic light (chopped at 210 Hz) on the device pixel. The system was calibrated using a Si reference cell for wavelengths of 300–1,100 nm and a germanium (Ge) reference cell for the 1,100–1,400 nm range. For bias illumination, a filtered 150 W white lamp (with 550-nm or 850-nm optical filters) is employed to measure the bottom and top subcells of the tandem devices, respectively.

SEM and EDX spectroscopy

Field-emission SEM (TESCAN, MIRA3) with a beam energy of 5 kV was applied for surface morphology characterization. The EDX mapping was achieved through an EDX spectroscopy spectrometer (Oxford Ultim Extreme) with Aztec HKL standard.

Scanning transmission electron microscopy (STEM)

The high-angle annular dark-field STEM images and the elemental EDX mapping were obtained at 200 kV in a JEOL JEM-NEOARM transmission electron microscope (JEOL).

AFM and C-AFM measurements

AFM and C-AFM measurements were performed using a Bruker Innova system under ambient conditions. The difference is that C-AFM employed a Cr/Au -coated probe (MikroMasch, HQ: NSC15; spring constant $k = 40\text{ N m}^{-1}$; frequency $f_0 = 325\text{ kHz}$). A high-precision programmable d.c. power supply (LBTEK, SPD3303X) was used to provide a precise d.c. bias to the sample. The sample was connected to a current amplifier (FEMTO, DLPCA-200) via a coaxial clip (JCOPTIX, LAE-C1-BNC) to ensure stable electrical connections.

KPFM measurement

KPFM images were scanned using a Pt/Ir-coated probe (Bruker, SCM-PIT-V2; $k = 3 \text{ N m}^{-1}$; frequency $f_0 = 70 \text{ kHz}$) at a 40-nm lift height.

The ASP of the KPFM image follows equation (1):

$$\text{ASP} = \Phi_{\text{sample}} - \Phi_{\text{tip}} \quad (1)$$

where Φ_{tip} and Φ_{sample} represent the tip and sample potentials, respectively.

XPS and UPS measurements

XPS was conducted on a Thermo Fisher Scientific K-Alpha. UPS was performed on a Thermo Fisher Scientific ESCALAB Xi⁺ with a UV light source of He I (21.2 eV) with a pass energy of 0.5 eV (dwell time 50 ms, bias voltage -6 V).

XRD measurement

Crystal structure analysis was performed by XRD (Rigaku SmartLab-3 kW) using Cu K α radiation.

Absorption and transmittance measurements

UV–visible–near-infrared absorption and transmittance spectra were acquired on a PerkinElmer Lambda 365 spectrophotometer across 300–1,100 nm.

FTIR spectrometer

FTIR analysis was performed using a Thermo Fisher Nicolet IS50 instrument.

Raman spectrum measurement

Raman spectroscopy measurements were performed using a HORIBA XploRA PLUS system with an excitation signal of 532 nm.

PL measurements

PL data were obtained using a Zolix Flex One Spectrometer with an excitation wavelength of 532 nm.

FLIM and TRPL measurements

The FLIM and TRPL experiments were performed on a PicoQuant MicroTime 200 system. A 405-nm pulsed laser at 1 MHz was focused on the sample via an Olympus MPLFLN100 $\times$ objective (numerical aperture 0.9). Emission passed through a 430-nm long-pass filter and was recorded with a single-photon avalanche diode. A time-correlated single-photon counting detector processed the electrical signals.

TRPL data were analysed by fitting to a bi-exponential function in equation (2):

$$I_{\text{PL}}(t) = A_1 \exp^{-t/\tau_1} + A_2 \exp^{-t/\tau_2} \quad (2)$$

Average carrier lifetimes (τ_{ave}) are calculated using equation (3):

$$\tau_{\text{ave}} = (A_1 \tau_1^2 + A_2 \tau_2^2) / (A_1 \tau_1 + A_2 \tau_2) \quad (3)$$

where A_1 and A_2 represent the amplitude coefficients of the fast and slow decay components, while τ_1 and τ_2 represent their corresponding lifetimes.

TA spectroscopy measurement

TA measurements were conducted on a TA spectrometer (HARPIA, Light Conversion, 40 kHz, PHAROS) at a 1,030-nm central wavelength with $\sim 120 \text{ fs}$ pulse duration. A 470-nm pump light from an optical parametric amplifier (ORPHEUS twins, Light Conversion) excited the samples, and both beams were focused to $\sim 64 \mu\text{m}$ on the sample. The fitting parameters of the relaxation dynamics were extracted using equations (2) and (3).

TPC measurement

TPC measurements were conducted on the devices using a 405-nm laser diode for excitation. Under short-circuit conditions and without background illumination, the TPC response was excited by a 200- μs square pulse. Current signals were acquired on a Tektronix DPO3034 oscilloscope via the voltage drop over a 5- Ω series resistor. A d.c. bias was supplied through an MRF544 bipolar junction transistor configured in a common-collector configuration. The decay transients were fitted to an exponential function of equation (4):

$$I(t) = I_0 \exp^{-t/\tau} \quad (4)$$

TPV measurement

TPV measurements were performed using a 405-nm laser, with light intensity regulated by a waveform generator (Agilent 33500B) and monitored via a linear photodiode. A second 405-nm laser provided a small perturbation, with pulse intensity set to produce a voltage change $< 10 \text{ mV}$. The photovoltage decays were fitted by an exponential function of equation (5):

$$V(t) = V_0 \exp^{-t/\tau} \quad (5)$$

EIS measurement

EIS was conducted on a CHI760E analyser (A.C. Impedance) in the dark over 100 Hz to 1 MHz with an a.c. bias of -1.1 V .

Hall measurement

The Hall test used the HL9900 equipment from TOHO Technology and tested the resistivity, carrier concentration and so on of the samples through four probes.

Contact angle measurement

The water contact angle was measured on a KINO SL200KS and quantified using the Young–Laplace fitting method.

TOF-SIMS measurement

TOF-SIMS measurements were conducted using a ToF SIMS V system (ION-TOF GmbH) in the positive ion mode.

Electrostatic potential

Electrostatic potential calculations and geometry optimization were performed with the ORCA 6.1.1 software package at the B3LYP/def2-SVP for ground-state structure. The optimized wavefunction was subsequently analysed with Multiwfn_2026.1.12, which was employed to compute the and generate the corresponding mapped surfaces.

First-principles calculations

All first-principles DFT calculations were performed using the ABACUS (Atomic-orbital Based Ab-initio Computation at USTC) software package, while the W_{f} calculations were carried out using the abacus-test package to ensure consistent postprocessing and data extraction. For detailed methods, please refer to Supplementary Note 10.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data generated or analysed during this study are included in the Article and its Supplementary Information. Source data are provided with this paper.

References

- Li, H. & Zhang, W. Perovskite tandem solar cells: from fundamentals to commercial deployment. *Chem. Rev.* **120**, 9835–9950 (2020).

2. Wu, W. et al. Stable and uniform self-assembled organic diradical molecules for perovskite photovoltaics. *Science* **389**, 195–199 (2025).
3. Jia, L. et al. Efficient perovskite/silicon tandem with asymmetric self-assembly molecule. *Nature* **644**, 912–919 (2025).
4. Wang, Y. et al. Homogenized contact in all-perovskite tandems using tailored 2D perovskite. *Nature* **635**, 867–873 (2024).
5. Fu, S. et al. Piracetam shapes wide-bandgap perovskite crystals for scalable perovskite tandems. *Nat. Nanotechnol.* **20**, 764–771 (2025).
6. Yang, X. et al. Understanding and manipulating the crystallization of Sn-Pb perovskites for efficient all-perovskite tandem solar cells. *Nat. Photonics* **19**, 426–433 (2025).
7. Ying, Z. et al. Antisolvent seeding of self-assembled monolayers for flexible monolithic perovskite/Cu(In,Ga)Se₂ tandem solar cells. *Nat. Energy* **10**, 737–749 (2025).
8. Tian, L. et al. Divalent cation replacement strategy stabilizes wide-bandgap perovskite for Cu(In,Ga)Se₂ tandem solar cells. *Nat. Photonics* **19**, 479–485 (2025).
9. Pei, F. et al. Inhibiting defect passivation failure in perovskite for perovskite/Cu(In,Ga)Se₂ monolithic tandem solar cells with certified efficiency 27.35%. *Nat. Energy* **10**, 824–835 (2025).
10. Zhang, J. et al. Highly efficient narrow bandgap Cu(In,Ga)Se₂ solar cells with enhanced open circuit voltage for tandem application. *Nat. Commun.* **15**, 10365 (2024).
11. Zeng, L. et al. A review of perovskite/copper indium gallium selenide tandem solar cells. *Solar RRL* **8**, 2301059 (2024).
12. Han, Q. et al. High-performance perovskite/Cu(In,Ga)Se₂ monolithic tandem solar cells. *Science* **361**, 904–908 (2018).
13. Jošt, M. et al. 21.6%-efficient monolithic perovskite/Cu(In,Ga)Se₂ tandem solar cells with thin conformal hole transport layers for integration on rough bottom cell surfaces. *ACS Energy Lett.* **4**, 583–590 (2019).
14. Al-Ashouri, A. et al. Conformal monolayer contacts with lossless interfaces for perovskite single junction and monolithic tandem solar cells. *Energy Environ. Sci.* **12**, 3356–3369 (2019).
15. Jin, Y. et al. Recrystallizing sputtered NiO_x for improved hole extraction in perovskite/silicon tandem solar cells. *Adv. Energy Mater.* **15**, 2403911 (2025).
16. Zhang, L. et al. Reducing defects: a review of preparation, doping and surface modification of nickel oxide. *Coord. Chem. Rev.* **534**, 216563 (2025).
17. Kafedjiska, I. et al. Integration of rough RTP absorbers into CIGS-perovskite monolithic tandems by NiO_x:(Cu)+ SAM hole-transporting bi-layers. *Solar Energy Mater. Solar Cells* **254**, 112248 (2023).
18. Liu, S. et al. Buried interface molecular hybrid for inverted perovskite solar cells. *Nature* **632**, 536–542 (2024).
19. Tang, L. et al. All-round passivation strategy yield flexible perovskite/CuInGaSe₂ tandem solar cells with efficiency exceeding 26.5%. *Adv. Mater.* **36**, 2402480 (2024).
20. Zhang, Y. et al. Reduction of nonradiative recombination at perovskite/C₆₀ interface in inverted perovskite solar cells. *Adv. Mater.* **37**, 2500501 (2025).
21. Kong, T., Chen, W., Zhao, Z., Liu, Y. & Bi, D. Catechol-functionalized self-assembled monolayer for highly efficient Sn-Pb and all-perovskite tandem solar cells. *Adv. Mater.* **38**, e17139 (2025).
22. Ma, C. et al. Photovoltaically top-performing perovskite crystal facets. *Joule* **6**, 2626–2643 (2022).
23. Wang, H. et al. Pre-annealing treatment for high-efficiency perovskite solar cells via sequential deposition. *Joule* **6**, 2869–2884 (2022).
24. Zhao, L. et al. Chemical polishing of perovskite surface enhances photovoltaic performances. *J. Am. Chem. Soc.* **144**, 1700–1708 (2022).
25. Blume, R. et al. Structural and chemical properties of NiO_x thin films: oxygen vacancy formation in O₂ atmosphere. *ChemPhysChem* **24**, e202300231 (2023).
26. He, J. et al. Improved anchoring of self-assembled monolayer on hydroxylated NiO_x film surface for efficient and stable inverted perovskite solar cells. *Adv. Funct. Mater.* **35**, 2413104 (2025).
27. Yu, S. et al. Homogenized NiO_x nanoparticles for improved hole transport in inverted perovskite solar cells. *Science* **382**, 1399–1404 (2023).
28. Li, Z. et al. Stabilized hole-selective layer for high-performance inverted pin perovskite solar cells. *Science* **382**, 284–289 (2023).
29. Peng, Y. et al. Enlarging moment and regulating orientation of buried interfacial dipole for efficient inverted perovskite solar cells. *Nat. Commun.* **16**, 1252 (2025).
30. Yang, F. et al. Minimizing interfacial recombination in 1.8 eV triple-halide perovskites for 27.5% efficient all-perovskite tandems. *Adv. Mater.* **36**, 2307743 (2024).
31. Chen, H. et al. Regulating surface potential maximizes voltage in all-perovskite tandems. *Nature* **613**, 676–681 (2023).
32. Si, H. et al. Emerging conductive atomic force microscopy for metal halide perovskite materials and solar cells. *Adv. Energy Mater.* **10**, 1903922 (2020).
33. Velilla, E., Jaramillo, F. & Mora-Seró, I. High-throughput analysis of the ideality factor to evaluate the outdoor performance of perovskite solar minimodules. *Nat. Energy* **6**, 54–62 (2021).
34. Ahn, N. et al. Trapped charge-driven degradation of perovskite solar cells. *Nat. Commun.* **7**, 13422 (2016).
35. Al-Ashouri, A. et al. Monolithic perovskite/silicon tandem solar cell with >29% efficiency by enhanced hole extraction. *Science* **370**, 1300–1309 (2020).
36. Ruiz-Preciado, M. A. et al. Monolithic two-terminal perovskite/CIS tandem solar cells with efficiency approaching 25%. *ACS Energy Lett.* **7**, 2273–2281 (2022).
37. Zheng, J. et al. Efficient flexible monolithic perovskite-CIGS tandem solar cell on conductive steel substrate. *ACS Energy Lett.* **9**, 1545–1547 (2024).
38. Tian, W. et al. Inert interlayer secures flexible monolithic perovskite/CIGS tandem solar cells with efficiency beyond 21%. *ACS Energy Lett.* **10**, 562–568 (2025).
39. Jeong, I. et al. Flexible and lightweight perovskite/Cu(In,Ga)Se₂ tandem solar cells. *Joule* **9**, 101794 (2025).
40. Best Research-Cell Efficiency Chart (NLR, 2026); <https://www.nlr.gov/pv/cell-efficiency>
41. Zhou, J. et al. Homogenizing hole-selective contacts for centimeter-square flexible perovskite/Cu(In,Ga)Se₂ tandems. *Sci. Adv.* **11**, ead2781 (2025).
42. Geng, C. et al. Two-terminal perovskite/Cu(In,Ga)Se₂ tandems with conformal coatings based on commercial bottom cells with >26% efficiency. *Energy Environ. Sci.* **18**, 8907–8917 (2025).
43. Pei, F. et al. Perovskite/CIGS tandem solar cells with over 1000 h operational stability through interconnection stress relief. *J. Am. Chem. Soc.* **147**, 36815–36824 (2025).
44. Li, W. et al. Efficient perovskite/Cu(In,Ga)Se₂ tandem solar cells with a composite intermediate recombination layer. *Nat. Commun.* **17**, 711 (2026).
45. Zhang, S. et al. Crystallization suppression of mixed-halide intermediates for perovskite/Cu(In,Ga)Se₂ tandem solar cells with improved efficiency. *Nat. Energy* **11**, 547–557 (2026).
46. Jost, M. et al. Perovskite/CIGS tandem solar cells: from certified 24.2% toward 30% and beyond. *ACS Energy Lett.* **7**, 1298–1307 (2022).
47. Green, M. A. et al. Solar Cell Efficiency Tables (Version 67). *Prog. Photovolt. Res. Appl.* **34**, 482–496 (2026).

Acknowledgements

We acknowledge the facilities and technical assistance provided by the Core Facility of Wuhan University. We thank W. Fan from the Core Facility of Wuhan University for assistance with SEM measurements. We also acknowledge the Theoretical and Computational Chemistry Team at Scientific Compass (www.shiyanjia.com) for assistance with the theoretical calculations.

Author contributions

J.L. and X.X. provided overall guidance for the project, secured primary funding and offered critical input during experimental discussions and manuscript preparation. L.Z., W.W. and L.T. contributed equally to this work. They conceived and designed the experiments, fabricated the tandem devices and performed a detailed analysis of the measurement results. Z.L. assisted in the optimization of the WBG PSCs. Y.Z. supported the experimental analysis process. Y.G. and H.X. contributed to the enhancement of semi-transparent device performance. X.W. and G.C. contributed to efficiency certification funding and valuable discussions on data analysis and manuscript writing. S.W. and Z.X. conducted AFM-related characterization and provided expert guidance on data interpretation. T.W., W.L. and Y.L. carried out TAS and XRD measurements. Y.S. performed FLIM and TRPL characterization. J.M., R.S. and J.Z. assisted with PL, TPC and TPV measurements. J.W. supported data processing and contributed to manuscript polishing. X.G. executed the DFT calculations and managed the data collection. W.Z. provided strategic guidance on computational design and interpreted the computational results. S.Y. supervised the theoretical calculations and supported computational resources.

Funding

J.L. acknowledges financial support from the National Key Research and Development Program of China (grant no. 2024YFB4205300), the Fundamental Research Funds for the Central Universities (grant no. 2042026kf0010) and the start-up funding from Wuhan University. X.X. and J.L. acknowledge financial support from the Special Fund

for the ‘Dual Carbon’ Science and Technology Innovation of Jiangsu province (Industrial Prospect and Key Technology Research Program, BE2022021). S.W. acknowledges financial support from the Wuhan Natural Science Foundation (grant no. 2024040801020219). G.C. acknowledges financial support from the National Natural Science Foundation of China (grant no. 62475044) and the Natural Science Foundation of Fujian Province (grant no. 2025J010031).

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41560-026-02125-3>.

Correspondence and requests for materials should be addressed to Xiaomin Wang, Guilin Chen, Xudong Xiao or Jianmin Li.

Peer review information *Nature Energy* thanks Qi Chen, William Jo and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Peer reviewer reports are available.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© The Author(s), under exclusive licence to Springer Nature Limited 2026

Solar Cells Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form is intended for publication with all accepted papers reporting the characterization of photovoltaic devices and provides structure for consistency and transparency in reporting. Some list items might not apply to an individual manuscript, but all fields must be completed for clarity.

For further information on Nature Research policies, including our [data availability policy](#), see [Authors & Referees](#).

► Experimental design

Please check the following details are reported in the manuscript, and provide a brief description or explanation where applicable.

1. Dimensions

Area of the tested solar cells

☒ Yes
☐ No

0.0539 cm² and 1.0298 cm² aperture areas were applied in the independent certification of small- and large-area devices, while 0.0585 cm², 0.0545 cm², 0.0506 cm², and 0.0408 cm² aperture areas were applied in some laboratory measurements.

Explain why this information is not reported/not relevant.

Method used to determine the device area

☒ Yes
☐ No

Determined by aperture masks verified by a CNAS-accredited certification laboratory (NIMTT, China).

Explain why this information is not reported/not relevant.

2. Current-voltage characterization

Current density-voltage (J-V) plots in both forward and backward direction

☒ Yes
☐ No

Forward and backward scans are provided in the Manuscript (Figs. 4c-d) and Supplementary Information (Supplementary Figure 77 and Supplementary Figure 80).

Voltage scan conditions

☒ Yes
☐ No

-0.1 V to 1.27 V for single-junction devices and from -0.1 V to 1.85 V for tandem devices (0.1 V s⁻¹ scan speed, 20 mV voltage steps, and 1 ms delay time), which can be found in the Methods section of the Manuscript.

Explain why this information is not reported/not relevant.

Test environment

☒ Yes
☐ No

All devices were measured under ambient conditions (room temperature of ~25 °C with a relative humidity of 40-60%).

Explain why this information is not reported/not relevant.

Protocol for preconditioning of the device before its characterization

☐ Yes
☒ No

Provide a description of the protocol.

No preconditioning was applied before the measurements in this work.

Stability of the J-V characteristic

☒ Yes
☐ No

The stability of the J-V characteristic is verified with time evolution of the maximum power point; see Fig. 5c, Supplementary Figure 77 and Supplementary Figure 82.

Explain why this information is not reported/not relevant.

3. Hysteresis or any other unusual behaviour

Description of the unusual behaviour observed during the characterization

☒ Yes
☐ No

The photovoltaic parameters of tandem devices can be affected by the area of the aperture mask, as described in the 'Performance and Stability of Perovskite/CIGS Tandems' section.

Explain why this information is not reported/not relevant.

Related experimental data

☒ Yes
☐ No

See Supplementary Figures. 78-79 and Supplementary Table 9.

Explain why this information is not reported/not relevant.

4. Efficiency

External quantum efficiency (EQE) or incident photons to current efficiency (IPCE)

☒ Yes
☐ No

The EQE was determined with a QE/incident photon-to-electron conversion efficiency (IPCE) system (Enlitech, QE-R666, China) by focusing monochromatic light (chopped at 210 Hz) on the device pixel. The system was calibrated using a Si reference cell for wavelengths of 300-1,100 nm and a germanium (Ge) reference cell for the 1,100-1,400 nm range.

Explain why this information is not reported/not relevant.

A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator

☒ Yes
☐ No

The EQE curves of the tandem cells were tested to verify the validity of the J-V measurements. See Fig. 4e, Supplementary Figures. 73-75 and Supplementary Figure 79.

Explain why this information is not reported/not relevant.

For tandem solar cells, the bias illumination and bias voltage used for each subcell

☒ Yes
☐ No

For bias illumination, a filtered 150 W white lamp (with 550 nm or 850 nm optical filters) is employed to measure the bottom (CIGS) and top (perovskite) subcells of the tandem devices, respectively.

Explain why this information is not reported/not relevant.

5. Calibration

Light source and reference cell or sensor used for the characterization

☒ Yes
☐ No

The illumination was calibrated to AM 1.5G using Enlitech AAA solar simulator, SS109 X100R, China.

Explain why this information is not reported/not relevant.

Confirmation that the reference cell was calibrated and certified

☒ Yes
☐ No

The output intensity of the light source was calibrated using the short-circuit current of a standard monocrystalline silicon (Si) solar cell from Enli Tech.

Explain why this information is not reported/not relevant.

Calculation of spectral mismatch between the reference cell and the devices under test

☒ Yes
☐ No

Adjust the intensity of the solar simulator lamp to correct the spectral mismatch between the reference cell and the device under test.

Explain why this information is not reported/not relevant.

6. Mask/aperture

Size of the mask/aperture used during testing

☒ Yes
☐ No

0.0539 cm², 1.0298 cm², 0.0585 cm², 0.0545 cm², 0.0506 cm², 0.0408 cm².

Explain why this information is not reported/not relevant.

Variation of the measured short-circuit current density with the mask/aperture area

☒ Yes
☐ No

The measured short-circuit current density fluctuates slightly with the aperture mask area, as shown in Supplementary Figures. 78-79 and Supplementary Table 9.

Explain why this information is not reported/not relevant.

7. Performance certification

Identity of the independent certification laboratory that confirmed the photovoltaic performance

☒ Yes
☐ No

National Institute of Measurement and Testing Technology (NIMTT, China)

Explain why this information is not reported/not relevant.

A copy of any certificate(s)

☒ Yes
☐ No

The certification reports are provided in the Supplementary Figure 77 and Supplementary Figure 80.

Explain why this information is not reported/not relevant.

8. Statistics

Number of solar cells tested

☒ Yes
☐ No

Represented by the specific sample sizes in the corresponding figure legends, or stated in the text.

Explain why this information is not reported/not relevant.

Statistical analysis of the device performance

☒ Yes
☐ No

Fig. 1b, Fig. 3b, Fig. 4b, Fig. 5a, Fig. 5c and Supplementary Figures. 1-3, 6-8, 37, 59-69, 71, 73-74, 82.

Explain why this information is not reported/not relevant.

9. Long-term stability analysis

Type of analysis, bias conditions and environmental conditions

☒ Yes
☐ No

The N₂ storage stability (~25°C under ambient light), thermal stability at 70 °C under ambient light, and MPPT stability (continuous 1-sun illumination; N₂ atmosphere; ~25°C room temperature) of the unencapsulated tandem devices are discussed in Fig. 5 and Supplementary Figures 81-83.

Explain why this information is not reported/not relevant.